Key Climate Features Driving Corn Yield Across US Corn Belt

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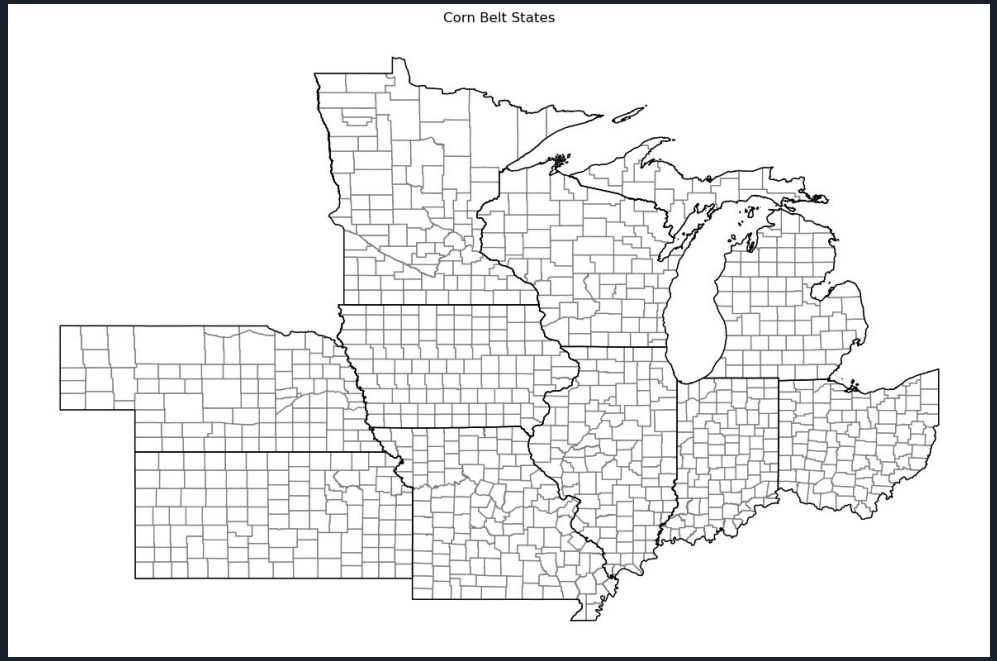
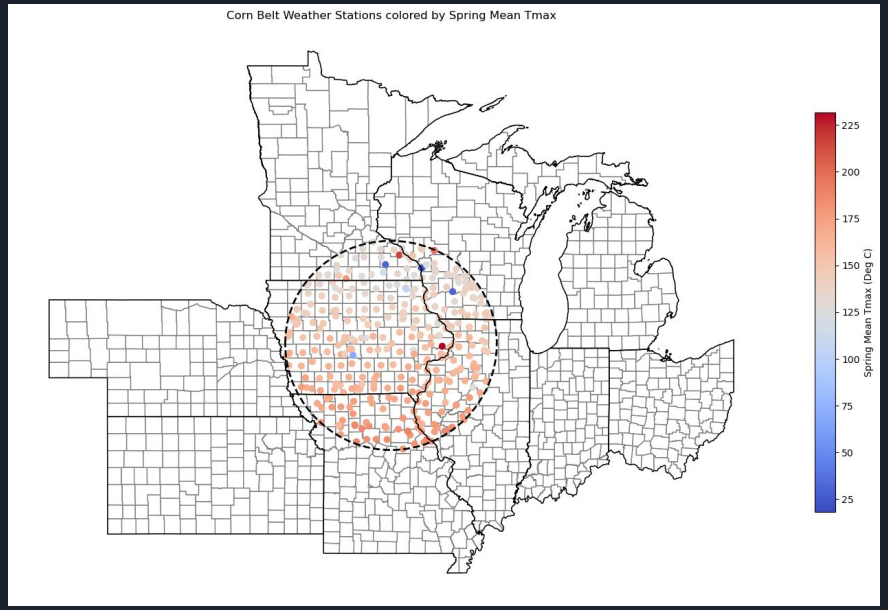
Intro & Objective

- Corn Belt is core of US corn production
- Yield is highly sensitive to climate variability
- Most models use fixed seasonal definitions, which limits interpretability
- **Research Question**: What predictors have the greatest impact on corn yield?



Methodology: 1950, 1958-2015 Data Creation (USDA annual yield + MET daily)

1. USDA Yield Data
    a. Load USDA county-level CSVs
    b. Detrend by fitting & subtracting linear year-yield trend per county
    c. Extract clean detrended_yield, county_name, year, state
2. MET weather data
    a. Load daily station CSVs, for corn-belt stations only
    b. Parse stations from GHCND inventory; filter to IDs present in MET data
    c. Spatial join each station to its county polygon (using geopandas & shapely)
    d. Tag each record with county_name, state, year
3. Merging
    a. Left-merge MET weather data → yield on (county_name, state, year)
    b. Drop any rows missing in deternded_yield
    c. Resulting merged table contains weather + detrended yield per station-year

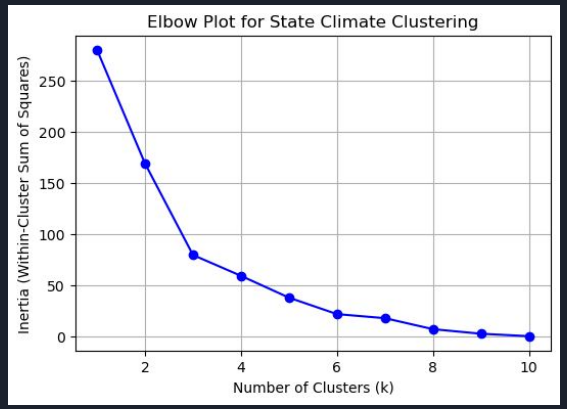


Methodology: Baseline Model

- Using only tmax, tmin, prcp, snow
- Previous attempts
  - Linear regression → R^2 ~ 0.0001
- Baseline model: random forest
  - R^2: 0.298

… So how could we improve accuracy and identify most important features for corn yield across the corn belt states?

Methodology: Clustering



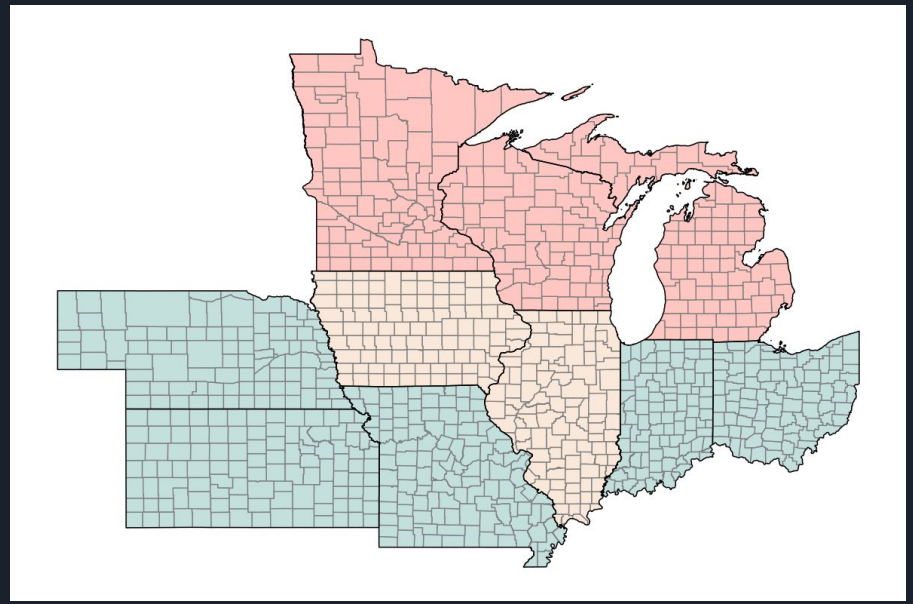
Data could form natural clusters

- Use k-means clustering to group data points with least variability
- Choosing # of clusters: use elbow method to choose 3 clusters for the scree plot

Cluster 0: MI, MN, WI

Cluster 1: IN, KS, MO, NE, OH

Cluster 2: IA, IL



Methodology: Feature Engineering

- Default features: tmax, tmin, prcp, snow
- Generate features for each season
  - Default seasons
    - Spring: 3/1-5/31
    - Summer: 6/1-8/31
    - Fall: 9/1-11/31
- For each season, generate **seasonal means, anomaly features, clear day thresholds**:
  - Relevant variables: precipitation, tmax, tmin, snow
  - Anomaly features
    - Thresholds
      - Each is based on percentile
      - Default: 90
    - Clear days threshold
      - Default: 5

Methodology: Hyperparameter tuning

- Attempted binary search algorithm
  - zigzag pattern in detrended yield → algorithm often converged to many local optima points
- Choose random search algorithm
  - For each cluster, randomly select season dates within certain seasonal ranges, restrict each season to be at least 14 days
  - Select best out of 200 iterations

Optimal Param	Spring	Summer	Fall	Anomaly percentiles	Clear
CLUSTER 0	3/17-4/3	5/18-8/21	10/1-11/7	'prcp': 82, 'tmax': 96, 'tmin': 99, 'snow': 99	3.14
CLUSTER 1	3/6-3/23	7/24-8/25	10/15-11/3	'prcp': 96, 'tmax': 85, 'tmin': 98, 'snow': 91	3.18
CLUSTER 2	3/13-3/28	7/6-8/31	11/4-11/29	'prcp': 80, 'tmax': 98, 'tmin': 94, 'snow': 93	2.69

Methodology: Model after tuning

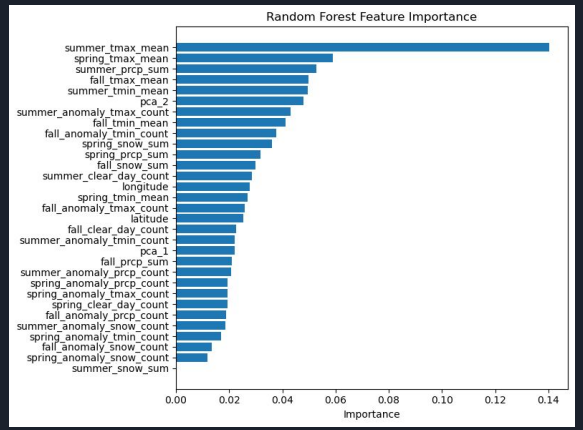
- **Simple RF model**
- **ML Pipelines:**
  - Stacked model (RF → XGB → MLP)
  - Residual stacking
- **More tuning**: Using PCA features, filtering NAs: using MICE, remove bias

R^2	Baseline (no tuning)	RF (w/ eng feat)	RF (w/ tuning)	Stacked	Residual Stacking	RF (after *pca*, *MICE*, *remove bias*)
CLUSTER 0	0.046	0.330	0.391	0.4205	0.6446	0.764
CLUSTER 1	-0.022	0.335	0.374	0.3867	0.4971	0.692
CLUSTER 2	0.079	0.561	0.568	0.5610	0.6284	0.772

Results: Consistent Yield Drivers across clusters

In spite of contrasting model behaviors, top predictors largely consistent across clusters, with slight rank-order shifts

Representative ex (Cluster 0)



- Summer maximum temperature is the strongest predictor in all clusters
- Spring & fall tmax reinforce seasonal heat sensitivity
- Heat anomaly frequency (extreme events) significantly impacts yield
- PCA component captures latent climate structure

Within the Corn Belt climate regime, seasonal heat intensity and extreme temperature events explain more variance in detrended yield than aggregate precipitation metrics.

Discussion

- Limitations
  - Perhaps the reason why RF worked better than our more complex models → tuned parameters using RF
  - → Highlights the significance of simple models & data extraction
  - Memory, computational restraint, server queue waits doesn't allow take in more variables such as ssrd, which could drastically improve accuracy
  - Precipitation effects may be underrepresented due to seasonal aggregation and interaction effects.
- Next steps
  - Tune parameters using more complex models → potentially help more complex models outperform RF
  - Extract more data in a more efficient manner, or server allocating more memory and CPU

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